

Characterization of urban distribution networks with light electric freight vehicles

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ABSTRACT

The massive incorporation of light electric freight vehicles (LEFV) in the last mile distribution has been identified as one of the alternatives that large Latin American cities shall include in their agenda for cleaner and more effective public policy in logistics operations. Thence, in this paper, a routing optimization model was used to evaluate the reduction in operational cost and GHG emissions when fueled cargo trucks or motorcycles are replaced by LEFV (i.e., e-trikes or e-bikes) in the last mile distribution in large Latin American cities. Results showed that for the case of a commercial area in Bogota, and two lines of business (i.e., parcel delivery and dry chain), the use of LEFV reduces more than 50% de operational cost and more than 95% the GHG emissions. These high reductions are due to the synergistic effect of switching from fuel to electricity and reducing the energy consumption of vehicles.

1. Introduction

The adoption of light electric freight vehicles (LEFV) is one of the basic measures for a more sustainable urban distribution possesses that faces several unresolved challenges, such as: i) identifying the appropriate mix of vehicle technology (fuel and electric powered) for a given available and future city mobility infrastructure (lanes, charging stations, and electricity) (Melander et al., 2022; Margaritis et al., 2016); ii) locate urban distribution centers to support the operation within the region of interest (Nordtømme et al., 2015), iii) improve the performance of LEFV to reduce operational costs (Zhang et al., 2019), and minimize air pollutant emissions, traffic congestion, and noise in the cities (Iwan et al., 2021; Melander et al., 2022); and iv) design the appropriate communication strategies of the already implemented experiences.

The United Nations forecasted that 70% of the population worldwide will live in urban areas by 2050. However, in Latin America, this figure is already 80%, and it is expected to rise to 86% by 2050 (UN, 2019; UN, 2020). This growing urban tendency and the magnitude of the logistical operations deployed to meet the needs of the population are imposing a new agenda for urban mobility, last mile distribution, air quality, and life quality of citizens, especially in large cities. Regarding air quality, 14% of the Global Green House Gases (GHG) emissions are generated by the transport sector (Lamb et al., 2021). It is clear that electro-mobility is an alternative for reducing the local and global emissions of conventionally fueled vehicles (Iwan et al., 2021; EUFAL, 2022).

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Despite the rapid technological development of electric vehicles, with more than seven million units circulating for everyday use, no growing tendency has been identified for the delivery and distribution industry (IEA, 2020), whose operation highly depends on the fleet type used. It is expected that electric-assisted freight bicycles and tricycles, henceforth just light electric freight vehicles (LEFV) show better performance in the last-mile distribution in terms of parking times, parking costs, ownership costs, delivery reliability, less severe collisions, and less air pollutant emissions (Sheth et al., 2019). Though their use also presents limitations, mainly in the size of the delivery zone, route length, demand density, topography, and constraint policies. e.g., the restriction of the use of LEFV in New York or the restriction of the use of e-trikes in certain bike lanes in Bogotá.

Following the European trend, Latin American governments are also investing in the development of bicycle-friendly city infrastructure, such as bike lanes, to further encourage this eco-friendly transportation mode, particularly in the most populated regions. (Nieves et al., 2019; Razvadauskas, 2018). For example, in 2020, a pilot test was carried out in a relatively dense residential and commercial area in Bogotá by LOGYCA and the City government, financed by the World Bank, and with the participation of different companies in the parcel and food delivery sector. This pilot tested the use of LEFV to support their last-mile distribution and was motivated by the need to i) develop strategies to reduce the logistic cost, ii) provide technical support for the formulation of public policy in logistics, iii) lead collaborative initiatives focused on the consolidation of cargo. It ran for six months in a collaborative cross-docking scenario.

1.1. Literature review on the use of LEFV

Although the study of incorporating electric vehicles in our society has been extensive, the adoption of LEFV has been much less revised (Melander et al., 2022). Motivators and barriers to the adoption of LEFV in logistics have been identified and classified into micro, meso, and macro levels (Melander et al., 2022). Environmental benefits are the main motivators for the adoption of LEFV, such as reductions in emissions of air pollutants (PM), GHG emissions, noise, and congestion (EUFAL, 2022; Choubassi et al., 2016; Melander et al., 2022). Financial issues usually appear at both sides of the coin: as motivators such as the total cost of ownership (TCO) reductions as well as barriers like the uncertainty of the financial return and the high costs of vehicle acquisition (Melander et al., 2022; TURBLOG ww, 2011) mainly because of the latent technology challenges.

The benefits of using electric vehicles (EV) in logistics operations can be enhanced by taking into account the particularities of different contexts (EUFAL, 2022). Hence, the European Electric Urban Freight and Logistics (EUFAL) project was launched in 2020 to provide a platform of exchange that provides tools for companies willing to incorporate electric vehicles into their fleets that are useful at different stages of the process. Previous European projects related to the use of electric vehicles in city logistics are summarized in Iwan et al. (2021). Their study is part of the EUFAL project and presents an efficiency study of electric freight vehicles in Szczecin, Poland, that shows that under a real-life environment, the autonomy and battery recharge are not significant limitations to the daily delivery operations. However, not much attention is given to LEFV's adoption in that project.

Studies on the incorporation of LEFV in last mile distribution appear in the last decade and are less frequent than for the other kinds of freight EV. One reason could be that they are not properly recognized or classified as freight vehicles yet. The vehicle classification in Margaritis et al. (2016) excludes bikes and trikes, arguing the lack of official specifications, and in the EUFAL's classification of electric vehicles (2020), the smallest vehicles are the light-duty electric vehicle with a capacity of up to 5 m³ and a gross weight up to 3500 kg. The assisted LEFV with a capacity of up to ~200 kg, and gross weight of <500 kg should be considered as a different category since energy requirements, capacities, and applications are quite different from other vehicles.

The financial benefits of using LEFV as an alternative to engine-powered vehicles for the last-mile distribution have been studied from different perspectives. For example, the introduction of different LEFV penetration rates in urban logistics activities with a maximum travel distance of 2 km in Porto, Portugal, showed up to 25% of reductions in external costs (Melo and Baptista, 2017), as well as 73% reductions in CO₂ emissions and a fleet replacement threshold of 10% of LEFV. In Barcelona and Valencia, Spain, using bicycles instead of motorcycles for delivery operations cost 14% less, and its maintenance costs represented 25% of that of a motorcycle (Navarro et al., 2016). Besides, LEFV are perceived to have costs reductions in other indirect costs such as taxes, insurance, storage, depreciation costs, parking time, and curb space occupancy (Melo and Baptista, 2017; Sheth et al., 2019). On the other hand, to compare operative costs against fueled vehicles, the cost estimation was distributed along the vehicle life expectancy in (Figliozzi et al., 2020), and initial investments and maintenance costs were considered in (Caggiani et al., 2021; Caggiani et al., 2021; Lee et al., 2019), and penalty costs were assigned to conventional vans in (Anderluh et al., 2017). Since much of the freight EV success depends on operational cost savings, approaches to optimize the specificities of vehicle routing and scheduling are still limited (Margaritis et al., 2016; Feng and Figliozzi, 2013).

The benefits of using EV in logistics operations cannot be widespread among the different lines of business, spaces, or contexts. Regarding lines of business, successful stories are mostly in grocery deliveries, home shopping deliveries, postal services, and perishable goods deliveries (Margaritis et al., 2016; SDA, 2020; EUFAL, 2022). As for spaces, many studies, such as TURBLOG and ENCLOSE projects (EUFAL), focus on megacities, historic downtowns or zones of high density.

Regarding the use of LEFV in urban distribution schemes, a recent systematic literature review reported 18 publications involving mathematical optimization since 2014 (Zazueta-Nassif et al., 2021). Those studies reported cases in North American, Asian, and European cities, whose urban contexts are different from those of Latin American cities. Figliozzi et al. (2020) focused on minimizing lifecycle emissions in Portland, finding emissions reductions of 4–6 times when using tricycles instead of conventional trucks, but recommended mixed fleets as the relative efficiency of conventional vehicles increases with load size and distance grow. Choubassi et al. (2016) compared the costs of fleet replacement for bikes and trikes in Austin and recommended the use of trikes under the assumption they can travel on bike lanes and locate the depot inside the demand area; Sheth et al. (2019) found that LEFV are a more

cost-effective option than trucks in Seattle and that the cost is sensible to the depot location. In the Asian continent stands out, Dalla Chiara et al. (2020) in Singapore, that studied other operational metrics such as travel distance and travel time, and Lee et al. (2019) in South Korea, that explored comparative costs of fleet replacement and estimated carbon emissions reductions. Several European studies also focus on time or distance-related optimization functions that consider the average speed and differentiated service times according to the type of vehicle (Zazueta-Nassif et al., 2021; Fikar et al., 2017; Gruber and Narayanan, 2019; Naumov and Starczewski, 2019). Other approaches take care of the risk for the biker (Osaba et al., 2018a; Osaba et al., 2018b), and CO₂ emissions (Akkad and Bányai, 2020; Figliozzi et al., 2020; Lee et al., 2019).

To assess the convenience of adopting LEFV as a substitution for diesel vehicles, specific and controlled scenarios are built to compare the performance of the various fleet compositions and sizes (Zazueta-Nassif et al., 2021). For example, the cost of three scenarios with different population densities was tested with different types of LEFV (Choubassi et al., 2016), showing the convenience of e-trikes in dense zones and the positive effect of the depot location within the delivery zone. Two scenarios with different cargo capacities and time windows constraints are used to compare the cost of using LEFV versus diesel vans (Tipagornwong and Figliozzi, 2014). Through an elasticity analysis, they found that their cost per mile is very sensitive to the driver cost and that the LEFV competitiveness is more sensitive to urban policies and on-street speed limits.

Also, three scenarios were compared in Antwerp (Arnold et al., 2018): a baseline with vans, the opportunity for customer self-pick-up, and distribution with cargo bikes and delivery points. As a result, authors suggest a combination of cargo bikes with self-pick-up points to improve quality of life since the use of cargo bikes improved the external costs –emissions, noise, and congestion– but not necessarily the internal costs.

Furthermore, four scenarios over the same route with different numbers of stops, number of parcels per stop, and distance from the distribution center to the delivery neighborhood in Seattle, US, showed better results in scenarios with high population density and with less than 2 km of travel distance (Sheth et al., 2019). Finally, two scenarios, one with only trucks and the other one with trucks plus cargo cycle and hubs, were explored under different demand and dwell times by Dalla Chiara et al. (2020). In the alternative scenario, consolidated loads were carried on trucks to the hub, and a predefined fleet of bikes performed the last-mile delivery. With a fixed demand density, sensitivity analyzes were made by varying the weighting factor of parcels, bikes' capacity, and dwell times. It showed that the last two factors were the most influential ones on travel times and distances, respectively.

1.2. About this work

From the former review, we observed that: (a) The exchange and dissemination of experiences with the particular conditions of each case help to generate confidence about the benefits of adopting LEFV in logistics operations and to counteract the barriers identified; (b) Cost savings are crucial for successful adoption of LEFV, so besides the further technological development of the vehicles themselves, the optimization of LEFV operation is needed according to the specific characteristics of the zone where they are going to be adopted (e.g., road network, depot location, demand density, etc.). However, we did not find studies that evaluate the bike lanes infrastructure of a city as a success factor for the adoption of LEFV; and (c) There is a growing interest in estimating the composition of a fleet considering electrified vehicles, but the inclusion of LEFV is quite rare.

Advancing previous works, the objective of this research is to identify the operational parameters within which LEFV can be incorporated in last-mile distribution so as to reduce the logistics costs of operation compared to traditional cargo vehicles such as trucks or motorcycles and, as a consequence, find the optimal level of fleet replacement.

Three specific questions were answered: (a) how does the adoption of LEFV affect the distribution cost? (b) what are the values of the relevant characteristics of a distribution network for which the incorporation of LEFV suites? and (c) how the transit of e-trikes on bike lanes affects the performance of the distribution operation? This last question was motivated by the fact that, due to their dimensions, e-trikes cannot travel on most of the bike lanes already available in cities.

The results in this study provide (a) technical but practical knowledge for decision makers on e-mobility last-mile delivery policy in large Latin American cities or any with similar population densities and bike lanes infrastructure, (b) guidelines to foster the mobilization of private sector resources to support the transition to eco-friendlier solutions for the delivery of goods in urban centers and (c) characterization of a distribution network that can be compared with any other urban zone to decide the convenience of the incorporation of LEFV as part of the urban distribution fleet of a company.

2. Methodology

Aiming at identifying the operational parameters within which LEFV can be incorporated in last-mile distribution, we selected a realistic but representative case. We selected a strategic region within Bogota city (Localidad Barrios Unidos) and obtained the real daily demand of two lines of business (parcel delivery and dry chain) within that urban zone. Then, we developed a model to predict the min-cost fleet mix and route to attend to that demand using real data of traffic and mobility infrastructure observed in that region. Finally, we compared operative cost and GHG emissions of the mixed fleet with respect to the use of 100% diesel-fueled vehicles (current case) under different scenarios of the location of the urban distribution center and the possibility of using the city's bike lanes for the operation of LEFV. Next, we describe each step of this general methodology.

2.1. The study region

We selected as the study region a commercial area within Bogota, Colombia, as a representative region of the largest cities in Latin America. In Colombia, nearly 50% of the energy demand and GHG emissions are due to the transport sector (Nieves, et al. 2019). Just in Bogota, 38.3% of the PM₁₀ emitted from mobile sources are caused by trucks, when these vehicles are only 3% of the city's fleet, while 38.5% of VOC and 23.3% of SO₂ are emitted by motorcycles, which are 21% of the vehicles in the city (SDA, 2020). It is expected that before 2030, Bogota will be the fifth Latin American megacity, after Mexico City, Buenos Aires, Sao Paulo, and Lima (Razvauskas, 2018), Bogota registered 611,000 daily bike trips in 2015, ranking second in Latin America in proportion of total daily trips on bikes (5%) after Rosario, Argentina (Ríos Flores et al., 2015). By 2019, Bogota continued reporting the highest values, with more than 880,000 daily bike trips, followed by Santiago de Chile and Mexico City (SDA, 2019). Furthermore, this city has more than 635 km of bike lanes (SDA, 2020), which is similar to Rio de Janeiro bike lanes infrastructure with 639 km of existing and planned bike lanes (Rio Prefeitura, 2015).

An almost flat area of approximately 17 km² was selected (Fig. 1). This area was utilized for a diversity of characteristics, like the presence of residential, commercial, and financial activity, higher commercial density than other areas of the city, and accessibility to numerous bike paths in the area.

Information on the available mobility infrastructure was obtained from google maps.

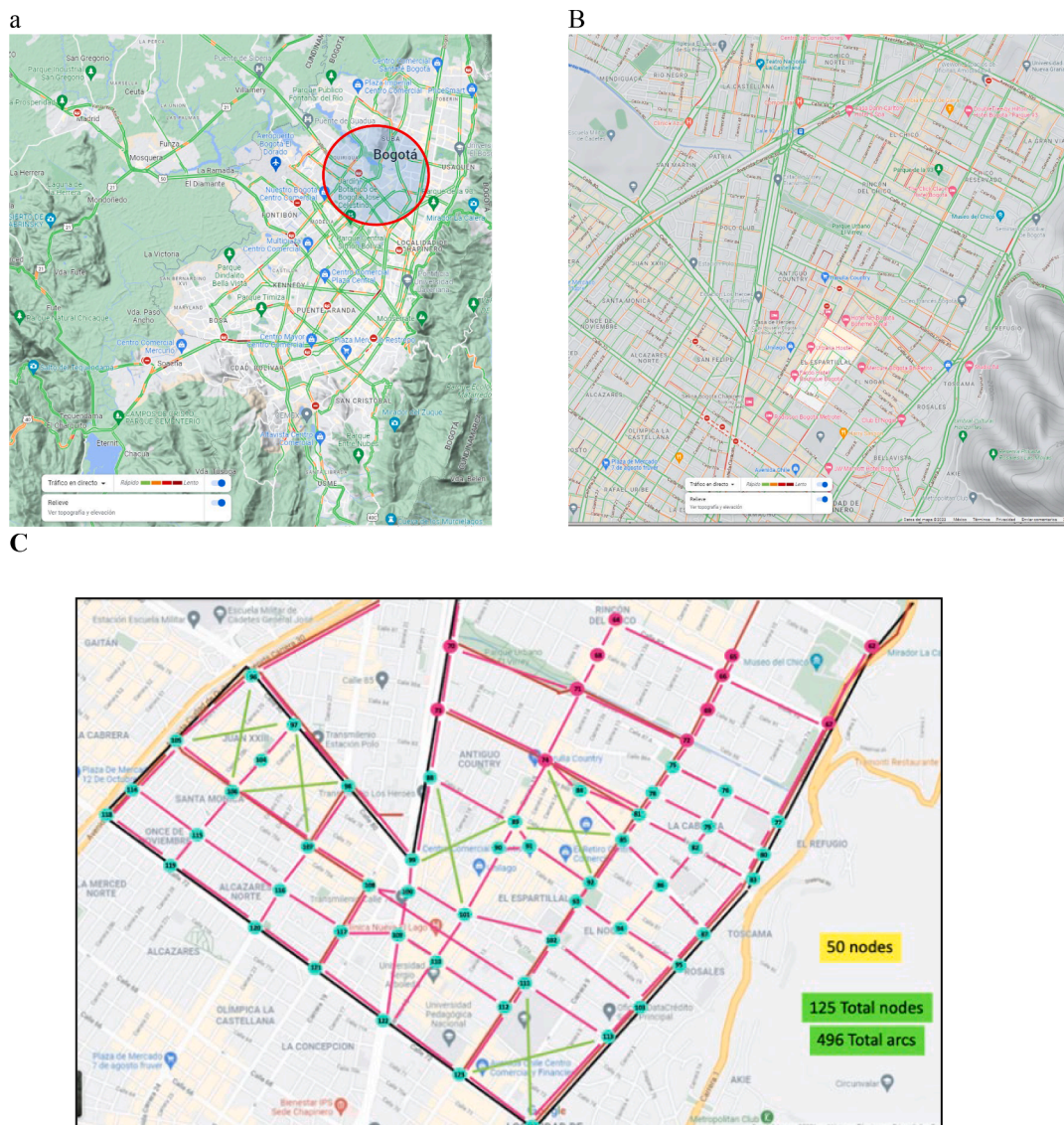


Fig. 1. Arcs definition example of about one third of the area modelled (Zazueta-Nassif et al., 2022).

2.2. Vehicles

The types of vehicles considered in this study were trucks, motorcycles, e-bikes, and e-trikes. Table 1 lists technical specifications and the operational parameters used for each of them. The meaning of the parameters is given in Table 2.

2.3. Last mile urban distribution in the study region

Bogotá, through the District's Secretary of Mobility, implemented in 2020 a pilot study that sought to promote alternatives to solve the current needs of last-mile distribution in the city (SDA, 2022). Several companies in the last-mile distribution industry participated. These companies used a collaborative cross-docking point located within the area of study. This work provided information on the location and demand of several lines of business within the industry of last-mile distribution, among which the more successful were dry chain (e.g., grocery home delivery) and parcel services. It also provided information on the cost of maintenance and labor force and on emissions using LEFV and traditional fueled vehicles (SDA, 2022).

2.4. Model description

We developed a mathematical model to select the minimum routing cost over a real road network. The optimization problem addressed in this work allows for identifying the best combination of vehicles from a heterogeneous fleet that distributes and satisfies from a transshipment (or depot) center the daily demand of an urban zone whose road network includes standard vehicle lanes and exclusive cycle lanes. The fleet includes internal combustion vehicles such as medium size trucks and motorcycles plus LEFV such as e-bikes and e-trikes. The objective is the minimization of the daily distribution cost, which is a key aspect of promoting replicability and further adoption of urban cycle logistics worldwide. The route solutions take care of time constraints, the delivery process, fleet capacity, and demand fulfillment. The solutions not just identify the sequence of customers to be visited but the actual route on the road network to be followed, in which arcs have attributes of speed and type of road (normal or bike lane). To allow the model to be used in lines of business whose packages are of different densities and dimensions, the demands and fleet capacities are described by volume and weight.

Prior to describing the model, Fig. 2 was developed to conceptually visualize the complete routing and distribution process. This example scenario represents a small network sample of nodes (28), with 25% of the arcs incorporating bike lanes and simulating bike lane networks in the city. In this figure, 15 client nodes were selected randomly (a number between 40% and 60% of the nodes). Two solution routes are depicted in blue and pink arrows. Initially, the merchandise arrives at the transshipment depot from the main depot, which is the company's warehouse. This displacement is not considered for the modeling because it would not be affected by the type of vehicle that makes the last-mile distribution.

The vehicle fleet that delivers the orders is in the transshipment depot, colored blue. Between this blue node and the network of nodes, a dummy or artificial station is proposed in the model to be used by those vehicles not required in the distribution process. Then, to enter the city network, all the vehicles in the fleet start in the transshipment depot, then they will travel to the dummy node, and only the vehicles that will serve clients continue to the network of nodes, which can only be accessed through node 1 or 2 in this example. Once there, each vehicle continues its route through the network and serves customers shown in pink or light blue. The purple dotted lines represent cycle lanes on the streets. Once the day's deliveries are finished, vehicles on the network must return to the transshipment depot, passing through node 1 or 2 and the dummy station again.

2.4.1. Assumptions

Regarding customers, it is assumed that a customer is served only by one vehicle, that the packages to be delivered have the same average size, and that the fixed and variable service times are the same for bicycles and motorcycles, while for the truck are higher.

For the vehicles, it is assumed that the fleet size of each type to be chosen is enough to satisfy the demand completely, that they all have the same average speed on the street, that only e-bikes can use the bike lanes, and that all streets allow the transit of all vehicles. In other words, the presence of bicycle lanes does not limit traffic on the street, but the cycling lane can have different speeds. The same speed for all the active vehicles on the street is assumed. Based on Hernandez (2020), the average speed during rush hours in the center of the city of Bogotá does not exceed 14 km/h when the traffic can be congested. So, the assumption is that the only way to increase the average speed would be with the use of bike lanes for applicable vehicles.

For the network, nodes do not represent clients' locations though the demand is allocated in the nodes. They are located at key points or intersections in the network to allow enough flexibility in the routing decisions. It is also assumed that multiple vehicles can

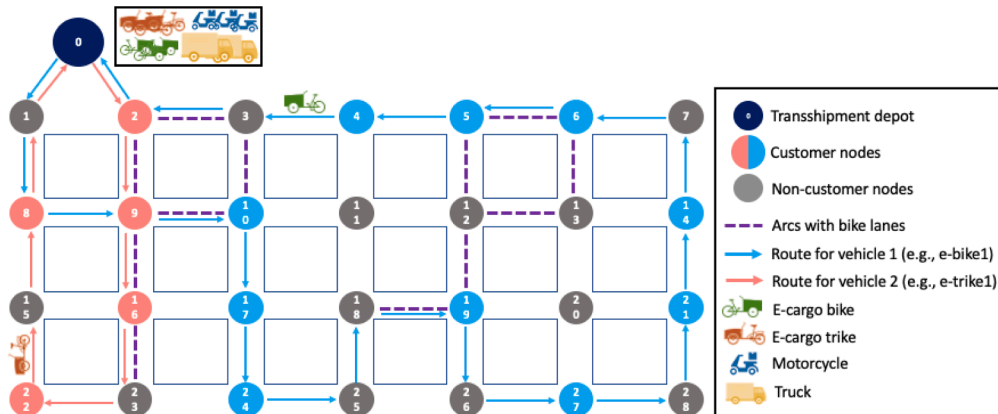
Table 1
Technical specification and parametric values used for the vehicles used in this study.

Parameter	$Cmax_k$	$Cmax_k$	$Dmax_k$	$Tmax_k$	Fuel	$Emax_k$	e_k	cf_k	cv_k	sa_k	stf_k	stv_k	SEC
Units	m^3	kg	km	hr		kWh or l	$Emax_k/km$	\$/day	\$/km	km/h	min	min	MJ/km
E-bike	0.196	100	30	8	Elec	0.56	0.006	22.54	0.00	8	6	5	0.47
E-trike	0.957	275	30	8	Elec	0.98	0.025	28.30	0.00	0	6	5	0.90
Moto	0.150	40	500	8	Gas	20.00	0.046	17.17	0.03	0	8	5	1.01
Truck	10.00	900	500	8	Die	95.00	0.125	47.48	0.22	0	12	8	8.67

Table 2

Notation used in the mathematical model.

Notation	Definition
Sets	
V	all nodes $i, j \in V, V = \{Vd \cup Vt \cup Vn\}$
$Vt \subset V$	transshipment depot node, $Vt = \{0\}$
$Vn \subset V$	all nodes for arc building, $Vn = \{1, \dots, N\}$
$Vu \subset V$	dummy node, $Vu = \{N+2\}$
$Vc \subset Vn$	customer nodes, $Vc = \{Vc^0, \dots, Vc^D\}$
$Va \subset Vn$	nodes connected to transshipment, $Va = \{1, \dots, h\}$
F	fleet of all vehicles available
$Feb \subset F$	all e-cargo bikes available, $Feb = \{1, \dots, k^{eb}\}$
$Fet \subset F$	all e-cargo trikes available, $Fet = \{k^{eb+1}, \dots, k^{eb+et}\}$
$Fm \subset F$	all motorcycles available, $Fm = \{k^{eb+et+1}, \dots, k^{eb+et+m}\}$
$Ft \subset F$	all trucks available, $Ft = \{k^{eb+et+m+1}, \dots, k^{eb+et+m+t}\}$
A	all arcs (i, j) in the network; $i, j \in V$
W	cargo features of each vehicle, $w \in W: W = \{1: \text{weight}, 2: \text{volume}\}$
Parameters	
r_{ijk}	binary parameter with a value of 1 if vehicle k is allowed to go through arc (i, j)
b_{ij}	binary parameter with a value of 1 if arc (i, j) has a bike lane
l_{ij}	length of arc (i, j) [km]
d_i	demand of node i [number of parcels]
cv_k	variable cost of vehicle k [\$/km]
cf_k	daily fixed costs for vehicle k [\$/h]
p_w	average parcel dimension [kg, m ³]
$Cmax_{kw}$	max cargo capacity for vehicle k in feature w [kg, m ³]
$Dmax_k$	max daily route distance for vehicle k [km]
$Tmax_k$	max shift time per day for vehicle k [h]
$Emax_k$	max energy capacity for vehicle k [kWh, l]
e_k	energy consumption rate for vehicle k , [kWh/km, l/km]
s_k	average speed for vehicle k [km/h]
sa_k	additional speed if vehicle k goes through an arc with a bike lane [km/h]
stf_k	service time fixed for vehicle k [h]
stv_k	service time variable for vehicle k [h]
N	number of nodes in the network
D	number of customers
M	An arbitrary large number $M > N$
h	Number of nodes connected to transshipment
SEC	Specific energy consumption
Variables	
z	total operation cost [\$/h]
U_{ikw}	remaining cargo of vehicle k regarding feature w on arrival at node i [kg, m ³]
H_{ik}	sequence of vehicle k for visiting node i
X_{ijk}	binary variable with a value of 1 if vehicle k goes through arc (i, j)
O_{ik}	binary variable with a value of 1 if vehicle k serves customer at node i

**Fig. 2.** Conceptual representation of the network and visual interpretation of the solution for the example scenario.

pass through the same node and that the bike lane adds an additional speed to the e-bikes as they travel through them. Bike lanes do not allow e-trikes. There can be arcs with only bike lanes. It is assumed that an arc with a bike lane is aside from a street road, so every vehicle can traverse this arc. In terms of costs, it is assumed that the variable component of the travel cost is proportional to the distance traveled, and the fixed cost corresponds to the total daily cost of ownership and operation.

2.4.2. Mathematical model

Let the directed graph $G = (V, A)$, where V is the set of nodes i (i.e., all the potential visiting nodes in the road network, with $V_t \subset V = \{0\}$, being the depot node, V_c the subset of customer nodes, and a dummy node in $V_u \subset V$), and A the set of arcs (i.e., feasible street segments between nodes). Table 2 lists the sets, parameters, and variables used in the model. The set of nodes in V are selected from the road network considering urban and traffic conditions (e.g., maximum one stop per block, stops located at one end of the edge, etc.). The heterogeneous fleet is divided into subsets according to the type of vehicle into e-bikes, e-trikes, motorcycles, and truck sets. So, the arcs can be traversed depending on their type. Parameters r_{ijk} and b_{ij} determine if the pass is available for a given vehicle and if they are exclusively bike lanes, respectively.

The mathematical model proposed to solve this problem is formulated as follows:

$$\min z = \sum_{(i,j) \in A} \sum_{k \in F} c v_k l_{ij} X_{ijk} + \sum_{i \in V_u} \sum_{j \in V_u} \sum_{k \in F} c f_k X_{ijk} \quad (1)$$

s.t.

$$\sum_{j \in V_u} X_{ijk} = 1, \forall i \in V_t, \forall k \in F \quad (2)$$

$$\sum_{i \in V_u} X_{ijk} = 1, \forall j \in V_t, \forall k \in F \quad (3)$$

$$\sum_{j \in V_u} M X_{ijk} \geq \sum_{i \in V_c} O_{ik}, \forall i = V_u \in V, \forall k \in F \quad (4)$$

$$\sum_{i \in V_u} M X_{ijk} \geq \sum_{i \in V_c} O_{ik}, \forall j = V_u \in V, \forall k \in F \quad (5)$$

$$\sum_{k \in F} O_{ik} = 1, \forall i \in V_c \quad (6)$$

$$\sum_{i: (i,j) \in A} r_{ijk} X_{ijk} = \sum_{i: (i,j) \in A} X_{ijk}, \forall j: (i,j) \in A, \forall k \in F \quad (7)$$

$$\sum_{j: (i,j) \in A} X_{ijk} \geq O_{ik}, \forall i \in V_c, \forall k \in F \quad (8)$$

$$U_{ikw} \leq U_{ikw} - d_i p_w O_{ik} + M(1 - X_{ijk}), \forall i \in V_c, \forall j: (i,j) \in A, \forall k \in F, \forall w \quad (9)$$

$$\sum_{i: (i,j) \in A} \sum_{j: (i,j) \in A} (l_{ij} e_k X_{ijk}) \leq Emax_k, \forall k \in F \quad (10)$$

$$\sum_{i: (i,j) \in A} \sum_{j: (i,j) \in A} l_{ij} X_{ijk} \leq Dmax_k, \forall k \in F \quad (11)$$

$$\sum_{i: (i,j) \in A} \sum_{j: (i,j) \in A} (l_{ij} / (s_k + s a_k b_{ij})) X_{ijk} + \sum_{i \in V_c} (stf_k + stv_k d_i) O_{ik} \leq Tmax_k, \forall k \in F \quad (12)$$

$$H_{ik} - H_{jk} + (N) X_{ijk} \leq (N - 1), \forall j: (i,j) \in A, \forall k \in F \quad (13)$$

$$U_{ikw} \leq Cmax_k, \forall i \in V_n, \forall k \in F, \forall w \quad (14)$$

$$U_{ikw}, H_{ik} \geq 0, \forall (i,j) \in A, \forall t, \forall k \in F, \forall w \quad (15)$$

$$X_{ijk}, O_{ik} \in \{1, 0\}, \forall (i,j) \in A, \forall k \in F \quad (16)$$

The objective function (1) is a cost minimization function, where variable and fixed costs are considered. The fixed cost of each vehicle is added only when it leaves the dummy node to one of the connected nodes and enters the arc network. This symbolizes the activation of one of the vehicles in the fleet.

Constraints (2) and (3) ensure that all vehicles, without exception, exit the transshipment depot to the dummy node and eventually

return. Constraints (4) and (5) ensure that only the vehicles that serve at least one customer leave the dummy towards the node network through one of the connecting nodes between the transshipment node and the rest of the network. For these restrictions, an arbitrary number M is used that must be greater than the number of nodes in the network. The following restrictions refer to sequencing and customer service issues. Constraint (6) ensures that all customers are served by only one vehicle. Constraint (7) is the flow balance that indicates that each vehicle that enters a node leaves the same node towards its next destination. Constraint (8) allows more than one vehicle to pass through the same client node, even when it does not serve it, repeating the restriction for each customer node and each vehicle in the fleet.

The next four constraints are related to the capacity aspects of the model. Constraint (9) keeps track of the load that each vehicle has at a given node and how it changes when making a delivery. Constraint (10) limits the energy consumption for each vehicle used to the maximum energy allowed for each vehicle of the fleet. Constraint (11) limits the daily distance traveled per vehicle to the maximum allowed distance, while constraint (12) restricts the daily travel time per vehicle to the available time window. To obtain the daily travel time, the arc distance is divided by speed, which can be increased with an additional speed when bikes traverse a bicycle lane arc. Travel time is added only if vehicle k transits that arc (i, j) . On the other hand, the service time is calculated considering the fixed service time per vehicle and the variable time multiplied by the packet demand of that node. Service time is added only if vehicle k serves client node i .

Finally, constraint (13) avoids loops or sub-cycles in the routes of each vehicle, with a standard restriction of loop avoidance, constraint (14) limits the maximum load that each vehicle can have, and constraints (15) and (16) are domain equations for the decision variables.

2.5. Network and data description

Nodes of the model network were identified considering the identification of strategic points in the area: access to bike paths, intersections between main streets, main roads, and strategic connector nodes. 125 nodes were defined, including the transshipment-depot node. The construction and definition of the arcs were carried out considering straight connections between the nodes. All the joints between nodes were enabled to travel in both directions, creating two arcs per connection. In addition to the straight arcs marked with pink (in Fig. 2) that correspond to road segments and allow transit to the four types of vehicles, additional arcs are defined and marked in green, representing the routes or movements that light vehicles can perform but conventional trucks cannot, such as crossing parking lots or parks or taking shortcuts through narrow streets. These arcs in green color are enabled in two directions, and access is only allowed to e-bikes, e-trikes, and motorcycles. After this process, 496 arcs were defined. This is the same network used in Zazueta-Nassif et al. (2022).

Afterward, values for the relevant parameters to be used in the model testing were taken or based on the pilot test (SDA, 2022) which are summarized in Table 1. The average fixed and variables costs correspond to average wages and fuel costs, respectively, also observed in the pilot; $Tmax_k$ is the standard Colombian shift time, and $Emax_k$ is autonomy which is related to the capacity of the gas tank for fueled vehicles. For the binary parameters (r_{ijk}) and (b_{ij}) , it was identified which of them allowed access for each type of vehicle and which had bicycle lanes. 150 of the 496 arcs have a bicycle lane (30.2%). For the parameter (l_{ij}) , the Euclidean distance between the nodes of each arc was used, although for the diagonal arcs, 1.2 times the Euclidean distance was used to consider the time spent in small turns of movement and that travels are not completely straight. These three parametric values for each node can be found in the Appendix.

For the parameter (d_i) , between 10 and 65 nodes were randomly chosen to define as client nodes, depending on the conditions of the simulation to be analyzed. For the case of the distribution of dry chain products, 10 ± 5 clients were identified for a low demand density and 30 ± 5 clients for a high demand density. Regarding the distribution of parcel and courier products, 35 ± 5 clients were identified for a low demand density and 60 ± 5 clients for a high demand density. For the parameter (p_w) , average dimensions of 0.04 m^3 and 2 kg were considered for the dry chain's line of business, and 0.02 m^3 and 1 kg for the parcel delivery line of business. These values were obtained from Logyca's pilot test. Additionally, fixed costs (cf_k) consider factors such as drivers' wages and rental fees. Variable costs (cv_k) are defined as the cost per kilometer and include energy or fuel cost, and maintenance costs. These values were obtained with Logyca's pilot test data and independent research.

Regarding the (s_k) parameter, it is the average speed in the modeling area. The value of (sa_k) is the additional speed (to the speed (s_k)) allowed on the bicycle lanes, and only for the e-bikes, as the rest of the vehicles cannot access them. Finally, the values of the (stf_k) and (stv_k) consider that the modeling nodes represent the geographic location and its immediate surroundings. The difficulty of parking for conventional distribution trucks causes the parametric times for this vehicle to increase due to the congestion in the area. On the other hand, the accessibility to sidewalks for e-trikes and e-bikes makes the times for these LEFV shorter than those of motorcycles.

2.6. Computational implementation

This scenario of about 500 arcs and the optimization model were implemented in Python® for the ease of using libraries and the friendly implementation of this process of modeling and programming for different parametric conditions, and mostly, the practical and free use of this model and the code by the test pilot participating companies. The open-source software suite for optimization OR-Tools was used with their library PYWRAPLP and the SAT solver. The full Python® code of the mathematical model can be found in the Appendix.

A convergence analysis was performed to identify and establish the time that the model would run in each tested scenario. The model was run for 15, 30, 60, and 120 min, each sample 3 times. Relative improvements in cost between 15-minute runs and 120-minute runs were of 7%. Variations in cost were caused by slight variations in route selection, but fleet selection remained the same. For these reasons, run times were set to 30 min in a 12 Gen Intel Core i712700KF-64 GB computer. Finally, regarding the dimension of the model, in the baseline scenario, there are 14,941 variables, considering the parametric values of 15 vehicles in the fleet, 125 nodes in the network, and two cargo parameters. On the other hand, the number of constraints varies depending on the density of the demand and the business line but ranges between 58,330 and 245,810 restrictions.

2.7. Model verification

An example scenario of the model was implemented and solved in GAMS/CPLEX 36.2.0 to verify its functionality. The full code as well as the input data of the small scenario can be found in the Appendix. The scenario has a network with 28 nodes, one node for the transshipment depot and two nodes connected to it. In addition, 15 clients were arbitrarily defined in different nodes of the network. The network has 96 total enabled arcs, considering the round trip between nodes. Of these, 24 had a bicycle lane for e-bikes. In addition, a fleet of eight vehicles was available: two e-bikes, two e-trikes, two motorcycles, and two trucks. After running the model, the optimal solution uses only two vehicles: one e-bike and one e-trike. The routes and the selected fleet are shown in the visual interpretation of Fig. 1. The customers served by the e-bike are shown in blue, and the customers served by the e-trike are in pink. It can be noticed that some client nodes are visited by both vehicles, such as 2, 8, and 9, but the established restrictions mean that they are only served by one of these vehicles. In addition, the results of the load, sequencing, service, and routing variables were analyzed in greater detail, and it is concluded that all behave according to the objective of the model, ensuring its functionality.

2.8. Quantification of GHG emissions

Starting from the results of the daily distance traveled by each vehicle under each scenario described previously, we calculated their respective GHG emissions in a life cycle assessment, which included emissions during fuel production (power source to fuel tank, WtT) and emissions during vehicle operation (from tank to wheel, TtW). The main GHG considered were CO₂, CH₄, and N₂O. The results were reported in terms of CO₂ equivalent (CO₂e), considering that CH₄ and N₂O have a global warming potential equivalent to 28 and 265 times that of CO₂, respectively (IPCC, 2014). Table 3 shows the emission factors considered per type of energy source, which were established by the USEPA (Zemo Partnership, 2022). Table 3 shows that 17 gCO₂e/MJ are emitted during the production of diesel (WtT analysis). When this fuel is used (TtW analysis), regardless of local conditions and the vehicle technology used, it emits 74.1 g/MJ of CO₂, and 3.9 mg/MJ of CH₄ and N₂O. Using the global warming potential stated above, these last emissions represent 75.24 g/MJ of CO₂e emitted to the atmosphere. Thus, in a life cycle analysis, 92.21 g of CO₂e are emitted per MJ of diesel used.

Similarly, Table 3 shows the emission factors for Gasoline and Electricity. As expected, Table 3 shows that no GHG are emitted during the use of any electric vehicles. However, the Colombian energy generation system emits 35 gCO₂e per every MJ of electricity produced (UPME, 2021).

Multiplying the emission factors reported in Table 3 by the density and energy content (LHV) of the fuel, the WtW emissions factors for diesel and gasoline can be expressed per L of fuel used. Finally, these emissions factors are multiplied by the specific energy consumption (SEC) of each vehicle technology (L/km) and the total daily distance traveled to obtain the total mass of CO₂e emitted during the day. We used the SEC observed in the last mile distribution in Bogotá, which were 23.02 and 3.18 L/100 km for diesel trucks and gasoline-fueled motorbikes, respectively. We also used 130 and 250 Wh/km for e-bikes and e-trikes, respectively.

3. Results

3.1. Design of experiments (DoE) for factors relevance

An initial 2⁵ factorial design was proposed, with five factors that were identified to have a potentially relevant impact on the behavior of the model. This DoE was preliminary discussed in Zazueta-Nassif et al. (2022). The line of business was chosen as the first factor having as levels, the two with better results in the pilot: Dry chain and Parcel service. The bike lanes factor refers to the proportion of bike lanes present in the road network, while average speed levels are set based on the speeds observed in the pilot. The speed factor can be interpreted as a simplified representation to compare rush versus flat hours, or two different congestion zones in a city, for example. The next factor is commercial density demand, measured in the number of customer nodes. However, the values of

Table 3
GHG emission factors for the operation of diesel, gasoline, and electric vehicles.

Energy source	Density	LHV	TtW	WtT				WtW
				CO ₂	CH ₄	N ₂ O	CO ₂ e	
	Kg/m ³	MJ/kg	g CO ₂ e /MJ	g/MJ	mg/MJ	mg/MJ	g CO ₂ e /MJ	g CO ₂ e /MJ
Diesel	826	45.61	17.0	74.1	3.9	3.9	75.24	92.21
Gasoline	740	42.72	18.8	69.3	25	8.0	72.12	90.94
Electricity	NA	NA	35	0	0	0	0	35.00

the two levels of commercial density defer between business lines, so it was decided to split the design into two 2^4 designs, one for each business line, as shown in Table 4. Finally, the depot location represents the two options of transshipment: the central level, as a representation of an exclusive distribution center for the business conveniently located in the middle of the demand zone, and the non-central option, as a way to represent a shared or collaborative distribution center, where its location does need to be conveniently in the center of the demand zone. In the experiments, the depot is located near the frontier of the demand zone.

Regarding the response variables that were selected as relevant to the study, five concepts were defined that help describe the decisions that the mathematical model makes and its implications in the logistics operation as in Zazueta-Nassif et al. (2022); starting with (1) the total cost of the operation, which can later be studied as the cost per package delivered. Also, (2) the number of activated units of each type of vehicle, (3) the average distance traveled by each category of vehicles, and (4) the average travel time of the two categories of vehicles. Finally, (5) the percentage of use of bicycle paths in the total route was also included, trying to identify to what extent the vehicles that access it favor the use of this infrastructure.

3.1.1. Scenario testing results

The mathematical model was run with the parameters of the 32 scenarios in random order. Each time, demand was randomly assigned to the demand nodes, with a uniform distribution [1,3] of packages per node. The response variables of each factor scheme can be viewed in Table 5. Experimental runs are identified following Yates notation (Montgomery, 2017). The response variables referring to the number of activated units of motorcycles and trucks are not added to the table since none was chosen in any of the 32 runs.

At first glance, it is motivating for the purposes of this research that the model heavily favors the use of LEFV in the analyzed schemes. For comparison purposes, the model was run again with the parameters of the 32 scenarios, but now disabling the LEFV of the fleet available. The resulting costs are presented in the last two columns of Table 5. The savings identified an average of 44% for the dry chain business line and 53% for the parcel service business when incorporating LEFV. It is observed that no motorcycles or trucks are chosen for delivery when LEFV are allowed, and when LEFV are taken out of the fleet, motorcycles are not used either, showing that it is the least efficient vehicle, as mentioned in previous research. The average percentage of use of bike lanes was 0.9% in scenarios where 10% of the arcs contained bike lanes and an average of 4.8% when 30% of the arcs contained bike lanes. The partial results shown so far were preliminarily presented in Zazueta-Nassif et al. (2022), and are necessary to discuss the following analyzes and results.

From Table 6, the average distance traveled per vehicle was significantly lower ($p < 0.001$) at the commercial density's high level for both lines of business: 24% for dry chain and 20% for parcel service, with average values of up to 20.4 km out of 30 km that were the maximum distances per route found. However, regarding travel time, the commercial density's effect was significant at about 5% for both lines of business. For the parcel service business, travel routes last up to 28% less than the 8 hr available per route. These values also provide evidence that the charging process can be done during the night without interfering with the delivery operation. This result is in line with Iwan's (2021) regarding autonomy and charging issues.

3.1.2. Analysis of variance

Eight analyzes of variance (ANOVAs) were run to identify which of the factors studied have significant impacts on the response variables. The five response variables and their first-order interactions were included as the object of study for each analysis carried out in Minitab, resulting in 10 factors for each ANOVA. The p-value results are summarized in Table 7. The complete eight ANOVA tables, together with the table of coefficients, are available in the Appendix.

The factor of commercial density or demand density is statistically the most relevant and statistically significant factor, with an average p-value of 0.013, for each response variable in both business lines. No interaction was identified as relevant and statistically significant. The depot location results are significant for the distribution cost and route time of the LEFV for the dry chain business.

The factor of average speed in the area was statistically relevant only for the variable average active time per vehicle for the Dry Chain line of business, being 15% lower at high speed (20 km/h). This factor had no statistical impact on any other variable, mainly because travel times were short compared to the eight hours available, and the additional speed of bikes when using the bike lanes can neutralize the effect of the different speeds in the rest of the network. The percentage of use of bike lanes was not significantly relevant in any case since the model was inclined to select e-trikes over any other vehicle, and the e-trikes couldn't access bike lanes. Finally, the factor of depot location has statistical relevance for the average distance traveled in both lines of business. Across all scenarios, the average distance traveled per vehicle increased by 29% (from 14.80 km to 19.09 km) when switching from a centralized depot to a non-centralized depot.

It is evident that the total cost of delivery will be directly influenced by the amount of demand to be delivered and the number of customer nodes to be visited. Therefore, being consistent with the scales of the experiment, the identified factor levels to minimize the

Table 4
Factors and levels for each DoE per business line: Dry chain and Parcel service.

Levels	A: Bike lanes	B: Speed	C: Commercial density	D: Depot location
Low	10%	15 km/h	Dry Chain: [8,14] Parcel Service: [25,35]	Central
High	30%	20 km/h	Dry Chain: [35,45] Parcel Service: [55,65]	Non-Central

Table 5
DoE results.

Line business	Yates order	Customers [nodes]	Demand [packages]	E-bikes selected	E-trikes selected	Average distance per LEFV [km]	Average time per LEFV [h]	Bike lanes usage % in route	Cost [USD \$]	Cost with no LEFV available [USD\$]	Saving with LEFV [%]	Reducion in GHG [%]
Dry Chain	(1)	10	16	0	1	15.897	2.480	0	28.36	51.03	44.41	97
	a	8	14	0	1	14.447	2.205	0	28.36	50.70	44.07	97
	b	12	16	0	1	16.378	2.266	0	28.37	51.13	44.53	96
	ab	10	14	0	1	13.469	1.935	0	28.35	50.48	43.8	97
	c	27	45	0	2	11.165	2.746	0	56.69	98.96	42.72	97
	bc	31	43	0	2	12.016	2.540	0	56.70	99.33	42.92	96
	ac	27	38	0	2	10.641	2.420	0	56.69	98.75	42.60	97
	abc	25	37	0	2	12.046	2.262	0	56.70	98.78	42.60	96
	d	11	17	0	1	21.916	2.982	0	28.39	52.37	45.79	96
	ad	14	19	0	1	19.564	3.021	0	28.38	52.09	45.52	97
	bd	9	15	0	1	21.539	2.411	0	28.39	52.73	46.17	97
	abd	9	12	0	1	18.216	1.996	0	28.37	51.54	44.95	97
	cd	33	42	0	2	13.273	2.792	0	56.71	100.51	43.58	97
	acd	30	43	0	2	15.851	2.991	0	56.73	100.40	43.50	96
	bcd	29	40	0	2	14.349	2.522	0	56.71	99.92	43.24	97
	abcd	31	42	0	2	17.833	2.789	0	56.74	99.39	42.91	96
Parcel Service	(1)	32	57	0	2	19.593	3.831	0	56.76	99.25	42.82	95
	a	30	50	1	1	16.981	6.587	14	50.93	99.91	49.03	97
	b	30	54	1	1	17.941	6.554	6	50.93	99.43	48.78	97
	ab	29	48	1	1	17.102	5.905	23	50.93	100.24	49.19	96
	c	61	113	0	3	15.858	4.386	0	85.09	201.36	57.74	97
	ac	58	112	0	3	13.761	4.209	0	85.07	201.65	57.82	97
	bc	59	106	0	3	14.33	3.845	0	85.07	197.64	56.96	98
	abc	61	118	0	3	15.293	4.232	0	85.08	200.44	57.55	97
	d	29	50	1	1	22.254	7.312	8	50.96	100.21	49.15	96
	ad	25	43	1	1	24.372	6.834	18	50.97	101.44	49.76	97
	bd	34	58	0	2	24.869	3.82	0	56.80	101.35	43.96	95
	abd	28	49	1	1	19.876	6.251	21	50.94	100.69	49.40	96
	bcd	60	110	0	3	18.206	4.153	0	85.12	203.42	58.16	97
	cd	56	102	0	3	17.462	4.172	0	85.11	204.49	58.38	97
	acd	64	119	0	3	16.968	4.636	0	85.10	205.00	58.49	96
	abcd	57	112	0	3	18.96	4.236	0	85.13	203.84	58.24	96

Table 6

Summary of distance and travel time per commercial density levels.

Business line	Distance per vehicle (or route) [km]		Travel time [h]	
	Commercial Density		Commercial Density	
	Low	High	Low	High
Dry Chain	$\bar{X}$: 17.68, σ = 3.16	$\bar{X}$: 13.40, σ = 2.48	$\bar{X}$: 2.41, σ = 0.41	$\bar{X}$: 2.63, σ = 0.28
ANOVA's p_value	0.007		0.043	
Parcel Service	$\bar{X}$: 20.37, σ = 3.13	$\bar{X}$: 16.35, σ = 1.85	$\bar{X}$: 5.89, σ = 1.34	$\bar{X}$: 4.23, σ = 0.22
ANOVA's p_value	<0.001		0.052	

Table 7

P-Value Results from ANOVAs.

Factors	Dry Chain Delivery				Parcel Service			
	Cost/ Parcel	Total Cost	LEFV Average distance	LEFV Average time	Cost/ Parcel	Total Cost	LEFV Average distance	LEFV Average time
Bike lanes %	0.174	1.000	0.320	0.169	0.409	0.252	0.370	0.372
Average Speed	0.136	0.576	0.479	0.009	0.321	0.999	0.930	0.572
Commercial Density	0.003	0.000	0.000	0.052	0.000	0.000	0.007	0.043
Depot Location	0.650	0.002	0.000	0.013	0.250	0.997	0.007	0.720

Table 8

Recommended Factor Levels for Cost per Parcel Delivered Minimization.

Line of business	Bike lanes [%]	Average speed [km/h]	Commercial density [customer nodes]	Transshipment depot location
Dry Chain	[10, 30]	[15, 20]	≥ 35	Central*
Parcel Service	[10, 30]	[15, 20]	≥ 65	Central**

*The cost per parcel is indifferent to the depot location. ** Only relevant for distance.

cost per parcel delivered for each line of business are listed in Table 8.

3.2. E-trikes over bike lanes

Because of the specific desire to study how bike lanes matter, new DoEs were run to further study what would happen to the use of bicycle lanes if the e-trike were redesigned to be able to traverse over the bicycle lanes. In this case, and based on the previous DoE analysis, the average speed was not included as a relevant factor. The dimensions of the e-trikes used in the pilot exceed the allowed dimensions to transit over bike lanes by approximately 20%. For this reason, access to bicycle lanes is enabled with a reduction of 20% of load capacity, calculating a new load capacity of 0.756 m³ and 270 kg, adhering to the provisions of the third Article of Resolution 160 of the Ministry of Transportation of Bogotá (Mayor's Office of Bogota, 2017).

Table 9 compares the results with previous scenarios, and Fig. 3 presents the effect of factor levels (bike lanes %, commercial density, and depot location) on relevant result variables (cost per parcel [\$], average distance per LEFV [km], and average time per LEFV [h]) when e-trikes are allowed in bicycle lanes. In these scenarios, more than 30% of the routes traverse the bicycle lanes in approximately a quarter of the time of each vehicle in transit. Full results for this DoE can be found in the Appendix.

Fig. 3 also shows that the cost per parcel is more sensitive to the three factors for the dry chain line of business than for Parcel service. In particular, the cost for dry chain decreases by 9% when the bike lanes percentage is at its high level (30%) while for parcel service, it decreases by 7%, and it is not affected by the depot location. In addition, it is observed that with larger densities, the length of the routes decreases and that for larger percentages of bike lanes in the road network, the average distances increase but the route times are shorter, which indicates that the routes tend to be slightly longer but faster, looking for using more the bike lanes. This can be seen in the average percentage of use of bike lanes. It was 13.6% in the low level (i.e., scenarios where 10% of the arcs contained bike

Table 9

Results when e-trikes are allowed to transit over bike lanes.

Line of business	Before: No e-trikes over bike lanes		E-trikes allowed over bike lanes	
	Bike lanes usage [%]	Average time [h]	Bike lanes usage [%]	Average time [h]
Dry chain	$\bar{X}$: 0	2.52	$\bar{X}$: 35, σ = 3.8, [27–41]	2.07
Parcel service	$\bar{X}$: 6, [0–23]	5.06	$\bar{X}$: 30.5, σ = 3.2, [24–35]	3.41

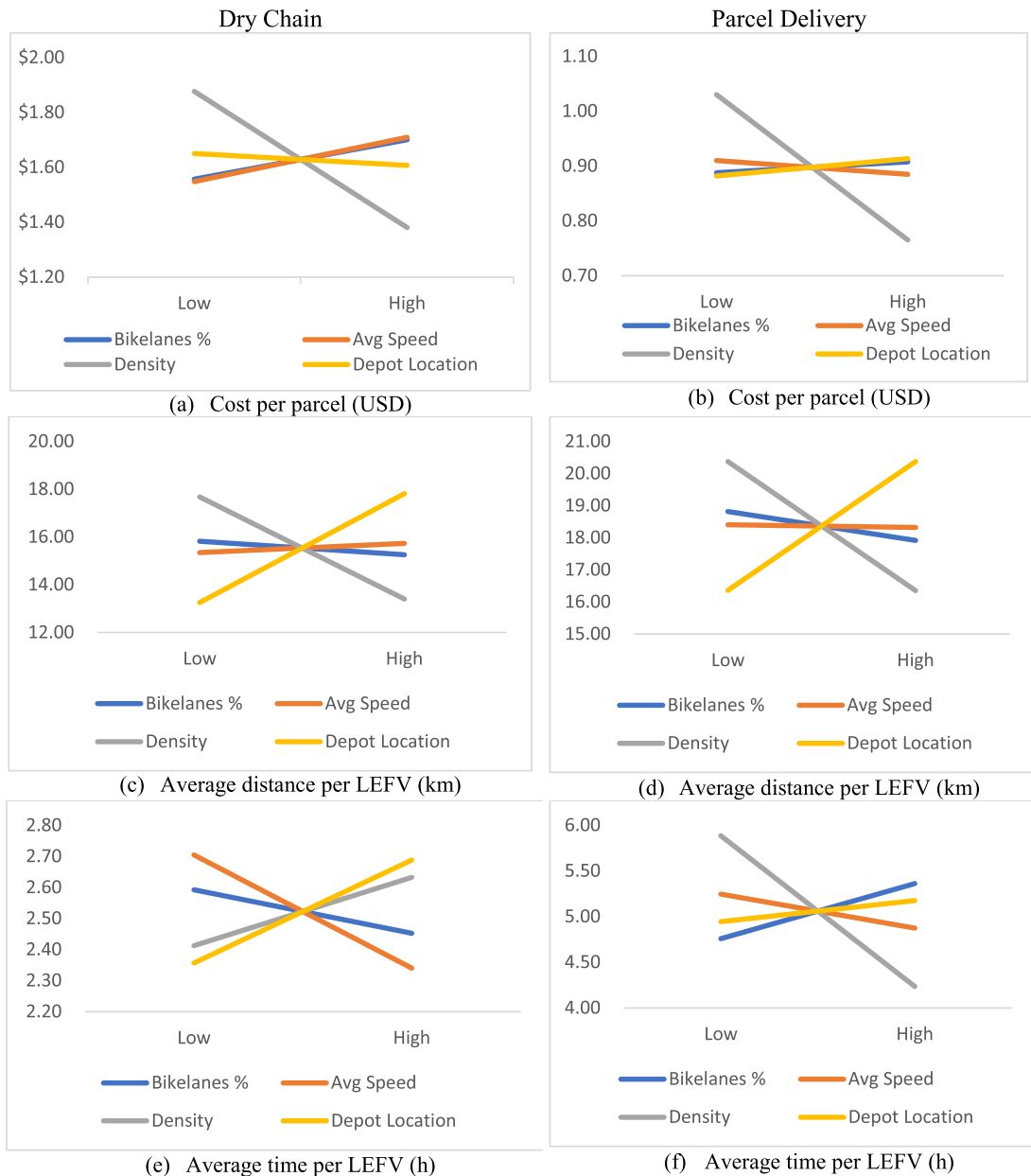


Fig. 3. Effects of relevant factors when e-trikes can travel over bike lanes for both lines of business.

lanes), and 35.6% in the high level (i.e., 30%). These values are much higher than the average usage of 0.9% and 4.8% for the two levels of this factor, respectively, when e-trikes were not allowed in bike lanes.

Another interesting result is observed in Table 9. The average time in which each vehicle is active is 2.07 h and 3.41 h for dry chain and parcel service business lines, respectively, reducing the average times by 18% and 32%, respectively, in reference to the scenarios where e-trikes couldn't get on bike lanes. Worth mentioning that none of the scenarios presents a time per vehicle greater than 4 h, despite having an 8-hour window to make the distribution.

3.3. The effect of commercial demand density

Given that commercial demand density was identified as the most significant factor among the ones studied, two additional experimental designs were designed and executed to further study this factor in each of the two lines of business. Three replicates of each level were run, going from 10 to 265 packages for dry chain and from 45 to 525 packages for parcel service, and allowing e-trikes travel on bike lanes. The limits are equivalent to 15 and 30 packages per km^2 approximately. Full results for this DoE can be found in

the Appendix. Relevant conclusions include that the costs for the three replicates or each level of demand number of packages are the same (i.e. vary only in cents), and that the amount delivered does not influence the use of the bike lanes, since the behavior is very similar to the one shown in Table 9.

Fig. 4 shows that one vehicle is needed for every 22 and 43 daily packages for dry chain and parcel service, respectively. This relationship remains linear, with $R^2 > 95\%$ even more than seven times the demand observed in the pilot test. From Fig. 4, it is also observed that in a few replicates, the solution includes the use of one truck or one e-bike, and e-trikes account for above 90% of the selected fleet on average, for both lines of business.

3.4. GHG emissions

As stated in the methodology section, in a life cycle analysis (WtW), 92.21, 90.94, and 35 gCO₂e (Table 3) are emitted to the atmosphere per MJ of energy used in diesel, gasoline, and electric-powered vehicles, respectively. However, when the typical vehicles' energy consumption is taken into consideration, for the case of last mile distribution in Bogota, the GHG emission becomes 799.8 gCO₂e/km for diesel-fueled trucks, 88.6 gCO₂e/km for gasoline motorcycles, 31.5 gCO₂e/km for e-trikes and 16.4 gCO₂e/km for e-bikes. We highlight that these values are estimates of average values of GHG emissions that hold only for the last mile distribution of parcel delivery and dry chain under the conditions observed in 2020 in the neighborhood Barrios Unidos of Bogota City.

Using previous emission factors, the total mass of GHG emitted per day was calculated for each scenario considered. Results are shown in Table 5 as the relative percentage of reduction compared to the respective scenario using traditional diesel-fueled trucks. It shows that in all cases, the full replacement of the actual diesel-fueled trucks by LEFV led to reductions above 95% in a country where 71% of the electric energy is generated by non-carbon sources of energy.

3.5. Discussion

Several findings deserve to be highlighted. Firstly, previous experiences and reviews have identified the most successful or potentially successful lines of business where the adoption of LEFV is appropriate (e.g., EUFAL 2022, Margaritis, et al. 2016). However, data-based comparisons among different lines of business are scarce, if any. Our results suggest different replacement rates of fueled vehicles for LEFV for the two lines of business studied as a function of the demand density. Secondly, the evaluated scenarios favor the use of LEFV for routes more than 20 km long. These results significantly extend the scope studied in Melo and Baptista (2017) and Sheth et al. (2019) with 2 km long routes. Next, motorcycles appear to be the most inefficient mode for the last mile delivery, and yet it is a mode that is widely used in Latin-American cities since they are perceived as very competitive when compared to other fuel-powered vehicles (Domecq and Sarmiento, 2015). Nonetheless, motorcycles have many cons, such as low capacity, high pollution levels, and very high accident rates. They could start to be competitive if the slope of the streets in the area is a relevant factor and probably if larger areas are analyzed. Though anyway, they share many of the barriers that are identified for LEFV. Finally, Results reinforce that, as in Iwan et al. (2021), the daily distribution operation is not very sensitive to battery charging issues (e.g., the current state of battery life, charging infrastructure, etc.) because route durations and energy consumptions are below daily or one-charge limits, so the recharging process can be done outside of operating hours, at least for the contexts studied.

4. Conclusions and further research

This paper studies the operating parameters under which a last-mile distribution company can introduce partially or totally LEFV as a replacement for the traditional fleet (i.e., fueled cargo trucks or motorcycles). Toward that end, we developed a routing optimization mathematical model to identify the best routes and the appropriate fleet mix to cover a daily last-mile delivery for cases with different demand, package density, vehicle technologies, and road network characterization (e.g., bike lane arcs), among other conditions. The model used the following assumptions, which require additional work to be relaxed, i.) the use of constant speed in all arcs of the network, ii.) the absence of road grade, iii) the presence of the distribution center (depot) within or on the limits of the study area, and iv) the assumption of one daily route per vehicle.

We used this model for the case of the last mile distribution of two lines of business (dry chain and parcel service) in a commercial neighborhood in Bogota where real data is available. Results reveal that e-trikes are the dominant mode over e-bikes, trucks, and above all, motorcycles, which were never part of the recommended fleets, and the last mile delivery costs can be cut in half when the fleet is primarily composed of LEFV. These results can be used as a reference to foster the use of LEFV in the last-mile distribution of Latin American urban spots. The availability of bike lanes infrastructure is only relevant when e-trikes are allowed to travel on them. This invites us to think about future research on e-trike designs that comply with dimensional regulations so that they can travel on bike lanes or to think about road infrastructure capable of supporting the transit of e-trikes.

It was also found that the average speed is not significantly relevant for cost, at least in the range studied (15–20 km/h), and depot location is a relevant factor to distance and total cost in the dry chain line of business. Therefore, the use of a cross-docking point favors the reduction of the distances traveled by cyclists on their LEFV. Only commercial demand density appears to be consistently significant not only to cost but to the other performance indicators, such as time and distance for both lines of business. These results could be extrapolated to similar contexts, with daily demands of up to 30 packages per km².

The replacement of light-duty trucks by LEFV for the last mile distribution in Bogota leads to >95% reductions in GHG emissions. This high reduction is due to the synergetic effect of i.) switching from diesel to electricity and ii.) reducing the energy consumption of vehicles by making them lighter and more efficient. Thus policymakers in the energy and environment areas should promote the use of

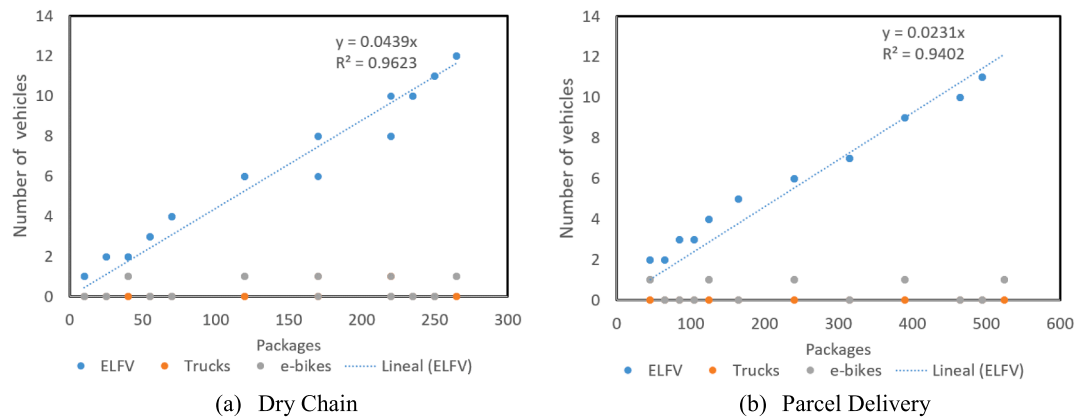


Fig. 4. Fleet size versus demand for both lines of business.

LEFV, especially in congested urban areas.

Directions of future research to further explore the benefits and conditions of using LEFV in the last-mile logistics operations are: to compare or evaluate differences in different city contexts (e.g., sloping streets, dynamic speeds, etc.) and other lines of business, to extend the range of the levels for the most significant factors, and to identify the conditions under which the use of motorcycles starts to be convenient. As for the speed, to incorporate the real values per arc in the network either dynamically or using the average values according to the official reports in the study area. About the mathematical model, it could be extended to consider multiple routes per LEFV on the same day. In addition, research is still pending on social, health, and safety issues for drivers to give operators a comprehensive view of the expectations of incorporating LEFV in their business and so contribute to the formulation of public policies in this regard.

CRediT authorship contribution statement

Jenny Díaz-Ramírez: Conceptualization, Methodology, Formal analysis, Writing – review & editing, Supervision, Project administration. **Sebastián Zazueta-Nassif:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization. **Roberto Galarza-Tamez:** Methodology, Software, Validation, Formal analysis, Investigation. **Daniel Prato-Sánchez:** Investigation. **Jose I. Huertas:** Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

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References

- Akkad, M.Z., Bányai, T., 2020. Multi-Objective Approach for Optimization of City Logistics Considering Energy Efficiency. *Sustainability* 1–23.
- Anderluh, A., Hemmelmayr, V., Nolz, P., 2017. Synchronizing vans and cargo bikes in a city distribution network. *CEJOR* 345–376.
- Arnold, F., Cardenas, I., Sörensen, K., Dewulf, W., 2018. Simulation of B2C e-commerce distribution in Antwerp using cargo. *Eur. Transp. Res. Rev.* 10 (2) <https://doi.org/10.1007/s12544-017-0272-6>.
- Caggiani, L., Colovic, A., Prencipe, L.P., Ottomaneli, M., 2021. A green logistics solution for last-mile deliveries considering e-vans and e-cargo bikes. *Transp. Res. Procedia* 75–82.
- Choubassi, C., Seedah, D., Jiang, N., Walton, M., 2016. Economic Analysis of Cargo Cycles for Urban Mail Delivery. *Transport. Res. Rec.: J. Transport. Res. Board* 102–110.

- Domecq, R., & Sarmiento, R. (2015). Logística urbana en motocicleta. *Revista Transporte y Territorio*, (13), 206–226. Available at: <https://www.redalyc.org/articulo.oa?id=333042522011>.
- EUFAL, 2022. EUFAL - Electric Urban Freight and Logistics. Retrieved February 10, 2023, from <https://www.eufal-project.eu/about.html>.
- Feng, W., Figliozzi, M., 2013. An economic and technological analysis of the key factors affecting the competitiveness of electric commercial vehicles: A case study from the USA market. *Transport. Res. Part C- Emerging Technol.* 135–145 <https://doi.org/10.1016/j.trc.2012.06.007>.
- Figliozzi, M., Saenz, J., Faulin, J., 2020. Minimization of urban freight distribution lifecycle CO₂e emissions: Results from an optimization model and a real-world case study. *Transp. Policy* 60–68.
- Fikar, C., Hirsch, P., Gronalt, M., 2017. A decision support system to investigate dynamic last-mile distribution facilitating cargo-bikes. *Int. J. Log. Res. Appl.* 39–58.
- Gruber, J., Narayanan, S., 2019. Travel Time Differences between Cargo Cycles and Cars in Commercial Transport Operations. *Transp. Res. Rec.* 623–637.
- IEA, 2020. Global EV Outlook 2020 Entering the decade of electric drive? IEA - International Energy Agency.
- IPCC, 2014. Cambio Climático 2014. Informe de síntesis. PNUMA, Grupo Intergubernamental de expertos sobre el cambio climático. [Equipo principal de redacción, R.K. Pachauri y L.A. Meyer (eds.)], Ginebra, Suiza. Retrieved from https://www.ipcc.ch/site/assets/uploads/2018/02/SYR_AR5_FINAL_full_es.pdf.
- Iwan, S., Nürnberg, M., Jedliński, M., Kijewska, K., 2021. Efficiency of light electric vehicles in last mile deliveries – Szczecin. *Sustainable Cities and Society* 103167, 1–10. <https://doi.org/10.1016/j.scs.2021.103167>.
- Lamb, W., Wiedmann, T., Pongratz, J., Andrew, R., Crippa, M., Olivier, J., Minx, J., 2021. A review of trends and drivers of greenhouse gas emissions by sector from 1990 to 2018. *Environ. Res. Lett.* 073005 <https://doi.org/10.1088/1748-9326/abee4e>.
- Lee, K., Chae, J., Kim, J., 2019. A Courier Service with Electric Bicycles in an Urban Area: The Case in Seoul. *Sustainability* 1255–1274.
- Margaritis, D., Anagnostopoulou, A., Tromaras, A., Boile, M., 2016. Electric commercial vehicles: Practical perspectives and future. *Res. Transp. Bus. Manag.* 4 (10), 4–10. <https://doi.org/10.1016/j.rtbm.2016.01.005>.
- Mayor's Office of Bogota, 2017. Resolución 160 de 2017 Ministerio de Transporte,. Retrieved 2021, from <https://www.alcaldiabogota.gov.co/sisjur/normas/Norma1.jsp?i=68085>.
- Melander, L., Nyquist-Magnusson, C., Wallström, H., 2022. Drivers for and barriers to electric freight vehicle adoption. *Transp. Res. D* 108 (103317), 1–13. <https://doi.org/10.1016/j.trd.2022.103317>.
- Melo, S., Baptista, P., 2017. Evaluating the impacts of using cargo cycles on urban logistics: integrating traffic, environmental and operational boundaries. *Eur. Transp. Res. Rev* 1–10.
- Montgomery, D., 2017. Design and Analysis of Experiments, 9th ed. Wiley, US.
- Naumov, V., Starczewski, J., 2019. Approach to simulations of goods deliveries with the use of cargo bicycles. In: *AIP Conference Proceedings*. AIP Publishing, Cracow, Poland, pp. 31–155.
- Navarro, C., Roca-Riu, M., Furió, S., Estrada, M., 2016. Designing new models for energy efficiency in urban freight transport for smart cities and its application to the Spanish case. *Transp. Res. Procedia* 314–324.
- Nieves, J., Aristizábal, A., Dyer, I., Báez, O., Ospina, D., 2019. Energy demand and greenhouse gas emissions analysis in Colombia: A LEAP model application. *Energy* 380–397. <https://doi.org/10.1016/j.energy.2018.12.051>.
- Nordtømme, M., Bjerkan, K., Sund, A., 2015. Barriers to urban freight policy implementation: The case of urban consolidation center in Oslo. *TransportPolicy* 44, 179–186. <https://doi.org/10.1016/j.tranpol.2015.08.005>.
- Osaba, E., Del Ser, J., Bilbao, M. N., Lopez-García, P., Nebro, A.J., 2018. Multi-Objective Design of Time-Constrained Bike Routes using Bio-inspired Meta-Heuristics. *BIOMA 2018: Bioinspired Optimization Methods and Their Applications*, 197–210.
- Osaba, E., Del Ser, J., Nebro, A. J., Laña, I., Bilbao, M.N., Sanchez-Medina, J.J., 2018. Multi-Objective Optimization of Bike Routes for Last-Mile Package Delivery with Drop-Offs. 2018 21st International Conference on Intelligent Transportation Systems (ITSC), 865–870.
- Razvadauskas, F.V., 2018. Megacities: Developing Country Domination. *Euromotor International*.
- Rio Prefeitura, 2015. Plano de Mobilidade Urbana Sustentável (PMUS) da cidade Rio de Janeiro. Rio Prefeitura, Transportes, Rio de Janeiro. Retrieved February 10, 2023, from <https://www.rio.rj.gov.br/documents/5450795/7289950/PMUS+RJ+-+P1+-+Plano+de+Execu%C3%A7%C3%A3o.pdf>.
- Ríos Flores, R.A., Taddia, A.P., Pardo, C., Lleras, N., 2015. Ciclo-inclusión en América Latina y el Caribe: Guía para impulsar el uso de la bicicleta. IDB - Inter-American Development Bank.
- SDA, 2019. Encuesta de Movilidad 2019, Bogotá. Secretaría Distrital de Movilidad, Bogota. Retrieved from https://www.movilidadbogota.gov.co/web/encuesta_de_movilidad_2019.
- SDA, 2020. Inventario de Emisiones de Bogotá. Contaminación Atmosférica 2018. Secretaría Distrital de Movilidad, Bogota.
- SDA, 2020. Nov. Muévete en bici por Bogotá. (Secretaría Distrital de Movilidad) Retrieved 06 2022, from <https://www.movilidadbogota.gov.co/web/muevete-en-bici-por-bogota>.
- SDA, 2022. BiciCarga. (Secretaría Distrital de Movilidad) Retrieved from <https://www.movilidadbogota.gov.co/web/galeria/bicicarga>.
- Sheth, M., Butrina, P., Goodchild, A., McCormack, E., 2019. Measuring delivery route cost trade-offs between electric-assist cargo bicycles and delivery trucks in dense urban areas. *Eur. Transp. Res. Rev.* 1–12.
- Tipagornwong, C., Figliozzi, M., 2014. Analysis of Competitiveness of Freight Tricycle Delivery Services in Urban Areas. *Transp. Res. Rec.* 76–84 <https://doi.org/10.3141/2410-09>.
- TURBLOG_ww, 2011. Handbook on Urban Logistics.
- UN, 2019. World Urbanization Prospects: The 2018 Revision. United Nations, New York: Department of Economic and Social Affairs - Population Division.
- UN, 2020. Global Issues: Population. (U. Nations, Producer) Retrieved from United Nations - Peace, dignity and equality on a healthy planet: <https://www.un.org/en/global-issues/population>.
- UPME, 2021. Factor de emisiones de la red de energía eléctrica en Colombia. UPME - Unidad de Planeación Minero Energética. Ministerio de Energía, Colombia. Retrieved from https://www1.upme.gov.co/siame/Documents/Calculo-FE-del-SIN/Documento_calculo_Cartilla_Factor_de_emision_2021.pdf.
- Zazueta-Nassif, S., Galarza-Tamez, R., Montemayor-Villarreal, V.P., Díaz-Ramírez, J., 2021. The Incorporation of Eco-Friendly Light Vehicles in Logistics Distribution Models: A Systematic Review. Proceedings of the International Conference on Industrial Engineering and Operations Management. November 3–5, Monterrey, Mexico, pp. 1686–1697.
- Zazueta-Nassif, S., Galarza-Tamez, R., Montemayor-Villarreal, V., Díaz-Ramírez, J., 2022. Evaluating the Benefits of Last-Mile Distribution with Light Electric Vehicles (LEV): A Case Study. Proceedings of the International Conferen. Stambul, Turkey: IEOM Society, pp. 3713–3725.
- Zemo Partnership, 2022. Zemo Partnership - Accelerating Transport to Zero Emissions. Retrieved Feb 10, 2023, from <https://www.zemo.org.uk/work-with-us/buses-coaches/low-emission-buses/well-to-tank.htm>.
- Zhang, Y., Jiang, Y., Rui, W., Thompson, R., 2019. Analyzing truck fleets' acceptance of alternative fuel freight vehicles in. *Renew. Energy* 134, 1148–1155. <https://doi.org/10.1016/j.renene.2018.09.016>.